On the Use of the `ubcdiss` Template

by

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OF THE REQUIREMENTS FOR THE DEGREE OF

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The following individuals certify that they have read, and recommend to the Faculty of Graduate and Postdoctoral Studies for acceptance, the thesis entitled:

On the Use of the `ubcdiss` Template

submitted by **Christophe Rouleau-Desrochers** in partial fulfillment of the requirements for the degree of **Master of Science in Forestry** in **Department of Forest & Conservation Sciences**.

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Abstract

This document provides brief instructions for using the `ubcdiss` class to write a UBC-conformant dissertation in LaTeX. This document is itself written using the `ubcdiss` class and is intended to serve as an example of writing a dissertation in LaTeX. This document has embedded Unique Resource Locators (URLS) and is intended to be viewed using a computer-based Portable Document Format (PDF) reader.

Note: Abstracts should generally try to avoid using acronyms.

Note: at University of British Columbia (UBC), both the Graduate and Postdoctoral Studies (GPS) Ph.D. defence programme and the Library's online submission system restricts abstracts to 350 words.

This document was typeset in `gpscopy` mode.

Lay Summary

The lay or public summary explains the key goals and contributions of the research/ scholarly work in terms that can be understood by the general public. It must not exceed 150 words in length.

Preface

At UBC, a preface may be required. Be sure to check the GPS guidelines as they may have specific content to be included.

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Glossary

This glossary uses the handy `acroynym` package to automatically maintain the glossary. It uses the package's `printonlyused` option to include only those acronyms explicitly referenced in the LaTeX source. To change how the acronyms are rendered, change the `\acsfont` definition in `diss.tex`.

GPS Graduate and Postdoctoral Studies

PDF Portable Document Format

URL Unique Resource Locator, used to describe a means for obtaining some resource on the world wide web

Acknowledgments

Thank those people who helped you.

Don't forget your parents or loved ones.

You may wish to acknowledge your funding sources.

Chapter 1

Wildchrokie

Introduction

1. Phenological shifts with climate change

   (a) Shifts in spring/autumn modify start and end of season

   (b) Early spring + delayed autumn should increase growth

   (c) This is important because carbon cycle models rely on this positive relationship

2. Maybe longer GS don't increase growth

   (a) Dow showed longer GS didn't increase growth locally

   (b) Cabon showed decoupling between GPP/Growth

3. Longer GS does not necessarily mean better conditions throughout the season

   (a) Longer seasons are not necessarily warmer

   (b) Early SOS does not increase GDD by much (Figure 1.1)

   (c) Water availability may constrain growth

   (d) Competition may offset extra days to grow especially for below canopy trees

4. Even with good conditions, trees may be internally constrained
    (a) Leaf phenology may be more plastic than growth
    (b) How plastic growth is depends on species and provenance
5. Local adaptation to inform growth with climate change
    (a) Trees from northern provenance may grow less with longer seasons
    (b) Budset is strongly locally adapted
    (c) Common gardens are important for climate change and tree growth
6. Answer how GS and growth relate using juvenile trees in a common garden
    (a) Disentangle GS length vs GS thermal conditions
    (b) Growth with tree rings, SOS and EOS with leaf phenology
    (c) Calendar season: days between SOS and EOS and thermal season: GDD between these days
    (d) Will look at different species from different provenances

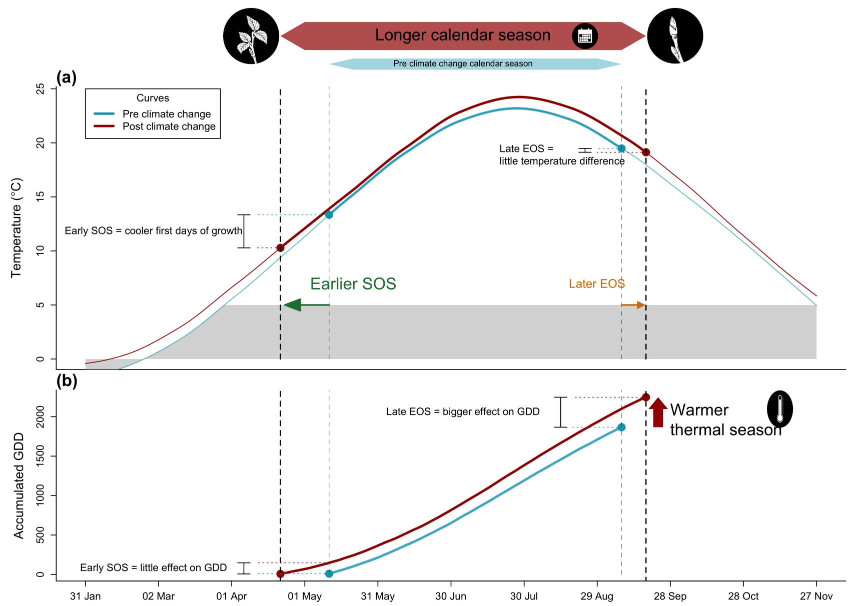

Figure 1.1: We show how shifting growing season length with climate change affects both the temperature trees experience during the growing season (a), and how much growing degree days (GDD) they accumulate (b). We used mean daily temperature data from the Boston airport Climate station for the smoothed temperature and GDD curves. We represent the changes in temperature before (1945-1965 in blue) and after the 1980s increase in climate warming (2005-2025 in red). With climate change, spring events (green arrow) have advanced and fall phenology (orange arrow) has delayed, but to much lesser extent (day advancements are fake and represented by the arrows), this lengthened the calendar growing season, but concurently changed the temperature trees experience early in their season. Growth onset earlier in the season means the first days of growth are cooler (a) resulting in little accumulation of GDD (b). Though shifts in EOS are weaker and the temperature experienced changed little (a), the GDD accumulation rate late in the season is higher than early in the season. Alongside warmer summer temperature, this results in higher GDD accumulation, warming their thermal seasons.

Results

1. Does tree growth increase with longer growing seasons?
2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?
4. How did growth change across years, and how did it relate to climate?

1. Summary basic information
   (a) Wildchrokie includes four deciduous tree species from four sites with leaf phenology data spaning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals.
   (b) Mean summer temperature were similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C)
   (c) The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018)
   (d) The earliest leafout day of year was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020). The earliest budset day of year was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018)
   (e) No drought for the three years, except for in 2020, where a moderate summer drought ramped up to a severe drought in September.
   (f) Annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018) (Figure S6).
2. Does tree growth increase with longer growing seasons?

(a) Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it.

(b) Longer thermal season increased growth for *B. papyrifera* ($\beta = 0.44$, UI_{90} [0.18, 0.7]), *B. populifolia* ($\beta = 0.22$, UI_{90} [0.07, 0.36]), and *B. alleghaniensis* ($\beta = 0.13$, UI_{90} [0.01, 0.24]), but *A. incana* had a weak effect, but trended in the same direction ($\beta = 0.07$, UI_{90} [-0.04, 0.17]) (Figure 1.2a and e) (TableS1).

(c) *A. incana* grew more with longer calendar seasons ($\beta = 0.05$, UI_{90} [0, 0.1]), but of the three species that responded to thermal season, only *B. papyrifera* also grew more with longer calendar seasons ($\beta = 0.18$, UI_{90} [0.05, 0.32]).

(d) *B. alleghaniensis* ($\beta = 0.03$, UI_{90} [-0.02, 0.09]), and *B. populifolia* ($\beta = 0.05$, UI_{90} [-0.02, 0.13]) trended in the same direction (Figure 1.2b and f) (Table S1).

(e) Results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table S4)

(f) On a scaled axis, both predictors were similarly predictive (FigureS7).

3. Did a later start and end of season increase growth?

   (a) A longer calendar season increased growth for two species, but only one grew more with an earlier start of season (*A. incana*: $\beta = -0.07$, UI_{90} [-0.14, -0.01]) (Figure 1.2c and g) (Table S1).

   (b) *B. alleghaniensis* which did not grow more with longer calendar season, grew less with an earlier start of season, unlike our prediction ($\beta = 0.1$, UI_{90} [0, 0.2]) (Figure 1.2c and g) (Table S1).

   (c) Three species grew more with a later end of season (*B. alleghaniensis*: $\beta = 0.12$, UI_{90} [0.05, 0.18], *B. populifolia*: $\beta = 0.1$, UI_{90} [0.02, 0.17]), with *B. papyrifera* ($\beta = 0.21$, UI_{90} [0.09, 0.33]) being the only one of those three that also grew more with longer calendar season (Figure 1.2d and h) (Table S1).

   (d) The estimates were unchanged when using the full (SOS $n = 239$ and $n = 230$) or restricted datasets ($n = 227$) (Figure S1a).

4. How did tree growth change across years, provenances and tree ids?

   (a) We did not find any clear effect of year on growth (Figure S5) and there is no clear differences of growth looking across species (Figure S2 and Figure S3)

   (b) Growth did not change across any provenances (Figure 1.3) (Table S2), and the variation across sites ($\sigma = 0.18$, UI_{90} [0.02, 0.5) and tree ids ($\sigma = 0.17$, UI_{90} [0.05, 0.27]) was similar

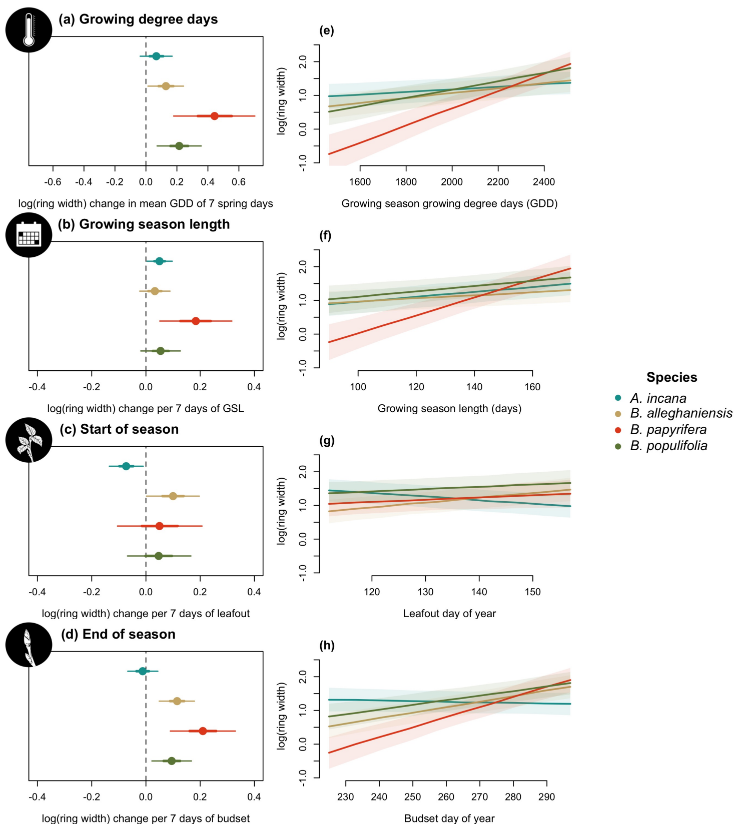

Figure 1.2: Model slope estimates representing ring width change as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and budset and the models estimates shows ring width change per 7 average spring days GDD (174.05) (a). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

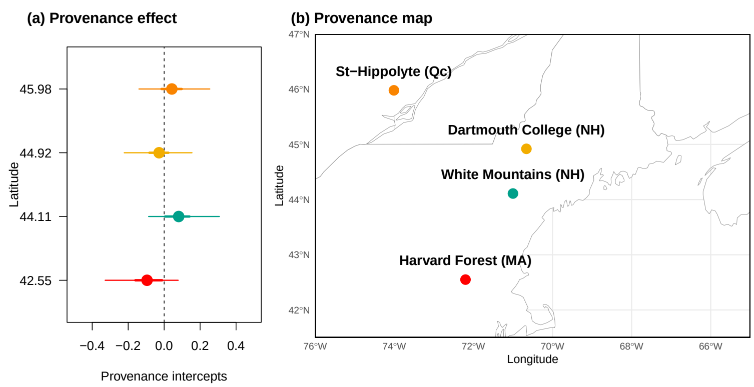

Figure 1.3: The effect of provenances on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor (effects were similar for the other predictors, see Table S2). The dots represent the mean of each provenance and the thick lines, the 50% and 90% uncertainty intervals (UI). The sites are color-coded from (b), the map showing their locations.

Discussion

1. Used 81 juvenile trees of 4 species from 4 provenances to understand if longer GS increased growth
    (a) We used annual growth with tree rings and leaf phenology for SOS and EOS
    (b) Longer GS increased growth for all species, but varied depending on the metric
    (c) Thermal season was a better predictor for growth than the calendar season
    (d) SOS and EOS trend direction changed across species
    (e) Little effect of provenance

A. Overall patterns

2. Why thermal season better predicts growth than the calendar season?
    (a) Thermal weights growth days better than calendar season
    (b) Cool spring days count for less for thermal season vs calendar season (Figure 1.1)
    (c) Stronger response to EOS than SOS, because temperature is warmer = more GDD (Figure 1.1)
3. SOS and EOS to disentangle the calendar season effect on growth
    (a) Early SOS didn't increase growth for most species (like Dow 2022)
    (b) Late EOS increased growth for most species; relates to GDD, less to GSL

B. Species specific patterns

4. Certain species

(a) *A. incana* grow more with GSL and earlier SOS: most growth happens in spring

(b) *A. incana* is ruderal, takes advantage of cool and short days in spring

(c) Contrary, *B. papyrifera* and *B. populifolia* driven by EOS: more reactive to warmer, late season temperature

(d) *B. alleghaniensis* SOS and EOS trends counterinteract = no response to GSL

5. No effect of provenance

   (a) Local adaption effect on growth is weak under current conditions

   (b) Little effect on budset too (Buonaiuto, unpublished)

C. Future directions, limitation and conclusion

6. Limitations

   (a) No extreme temperatures in models

   (b) Low sample size and not a fully crossed study design

   (c) Our trees would be below canopy in nature = light competition

7. What we did is important

   (a) Species trends differed, even if closely related

   (b) Carbon models that pool across species is a weakness

Supplemental results

Table S1: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
A. incana	0.07 [-0.04, 0.17]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
B. alleghaniensis	0.13 [0.01, 0.24]	0.03 [-0.02, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
B. papyrifera	0.44 [0.18, 0.70]	0.18 [0.05, 0.32]	0.05 [-0.10, 0.21]	0.21 [0.09, 0.33]
B. populifolia	0.21 [0.07, 0.36]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.17]	0.10 [0.02, 0.17]

Table S2: Site level intercept parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Site	GDD	GSL	SOS	EOS
Harvard Forest (MA)	-0.10 [-0.33, 0.08]	-0.10 [-0.34, 0.08]	-0.05 [-0.26, 0.11]	-0.09 [-0.33, 0.08]
White Mountains (NH)	0.08 [-0.09, 0.31]	0.07 [-0.10, 0.29]	0.08 [-0.07, 0.29]	0.08 [-0.10, 0.30]
Dartmouth College (NH)	-0.03 [-0.22, 0.15]	-0.05 [-0.26, 0.13]	-0.07 [-0.27, 0.09]	-0.04 [-0.24, 0.15]
St-Hippolyte (Qc)	0.04 [-0.14, 0.25]	0.03 [-0.16, 0.24]	0.02 [-0.15, 0.21]	0.05 [-0.14, 0.27]

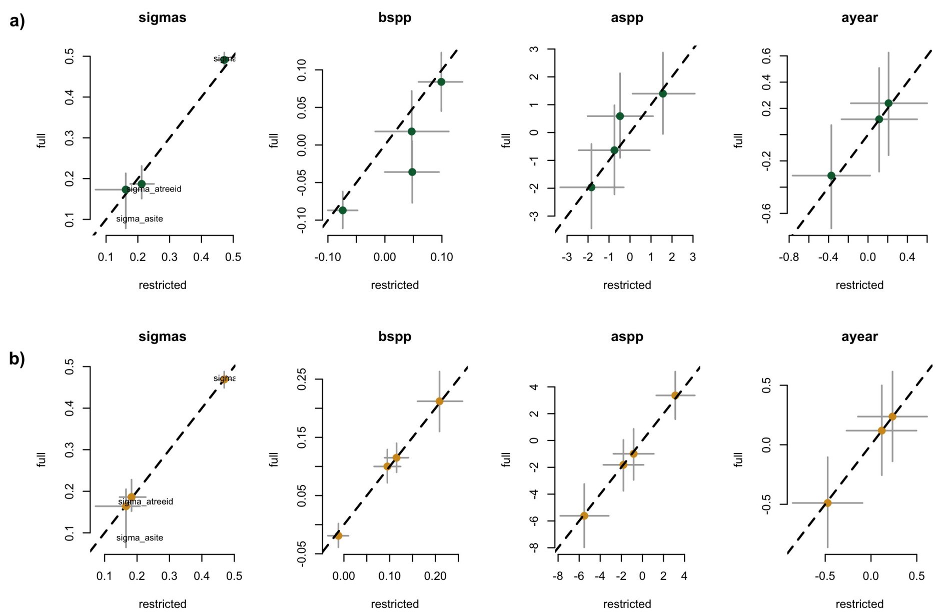

Figure S1: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor for Wildchrokie. Each pannel represent the different parameters used to fit the models.

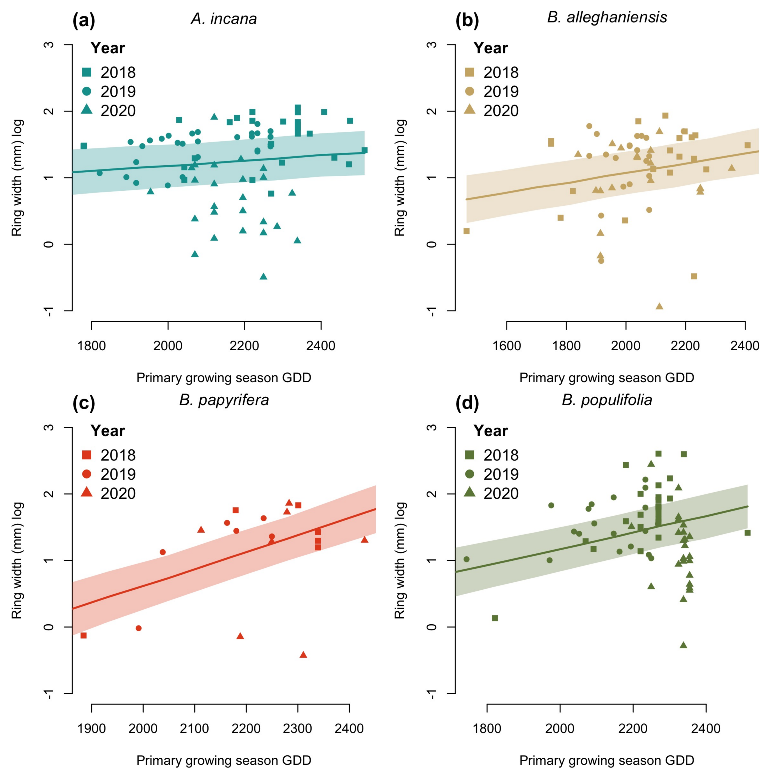

Figure S2: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

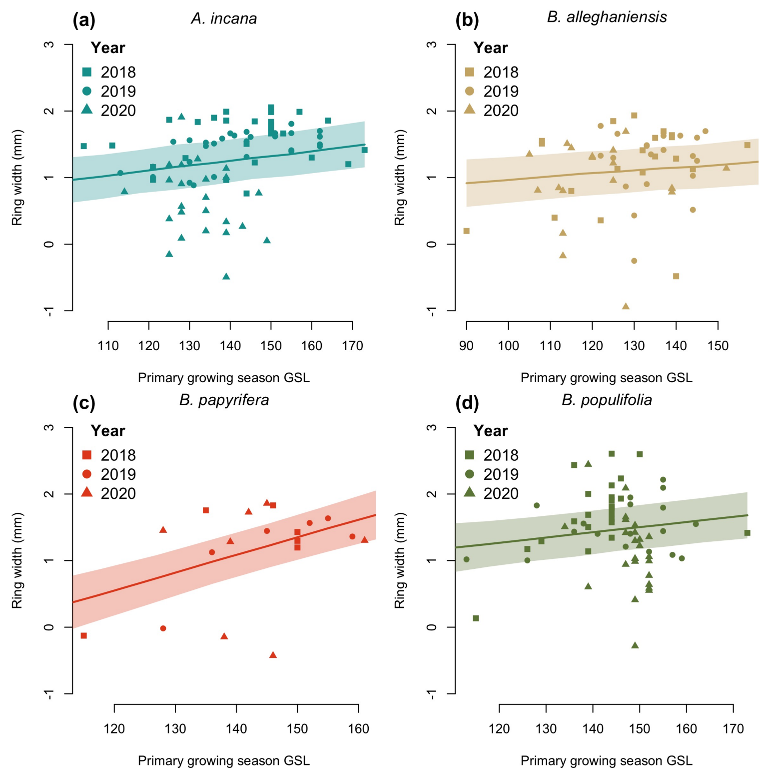

Figure S3: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

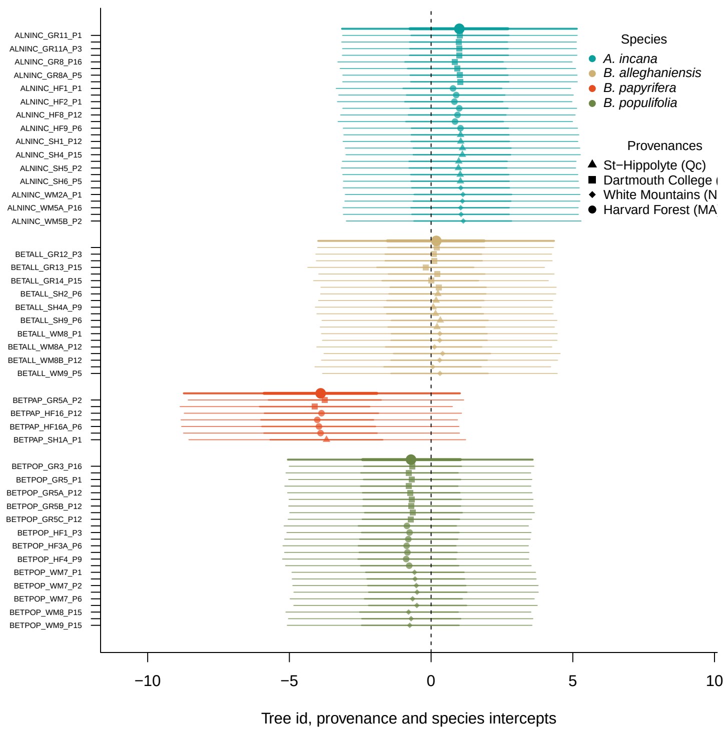

Figure S4: Intercept estimates for each group of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The bigger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

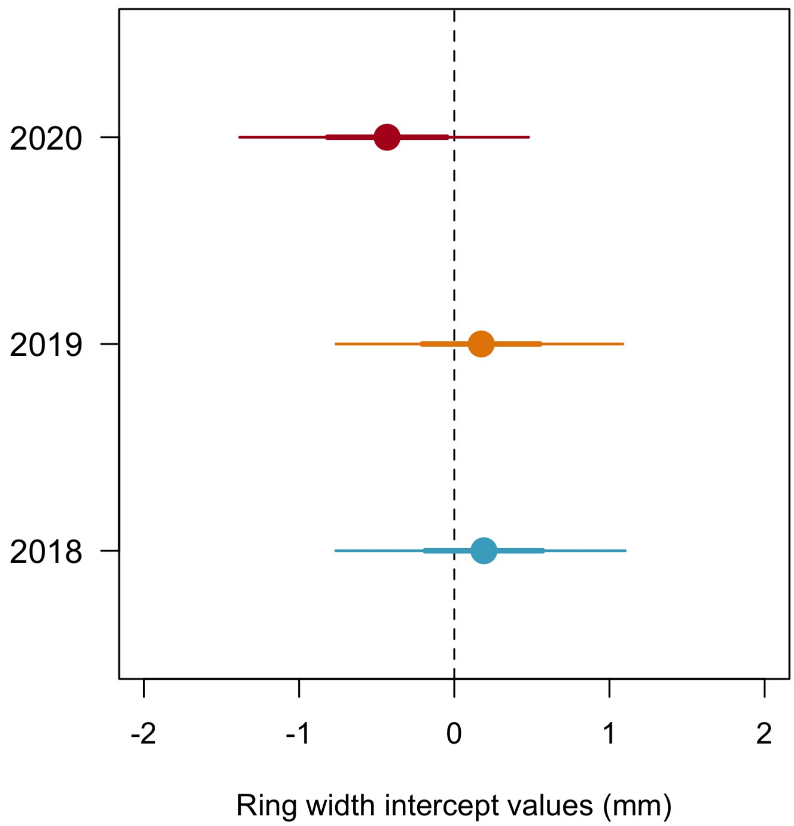

Figure S5: Model estimates for the year intercepts with the dots depicting the mean and the lines, the 50% and 90% uncertainty intervals (UI) and (b) using box plots of the empirical data. Though the trend is weak, 2020 is systematically the year with the lowest growth shown both as the mean of the model intercepts (a) and as the median of the raw data, for each species (b). The growth of 2018 and 2019 is similar.

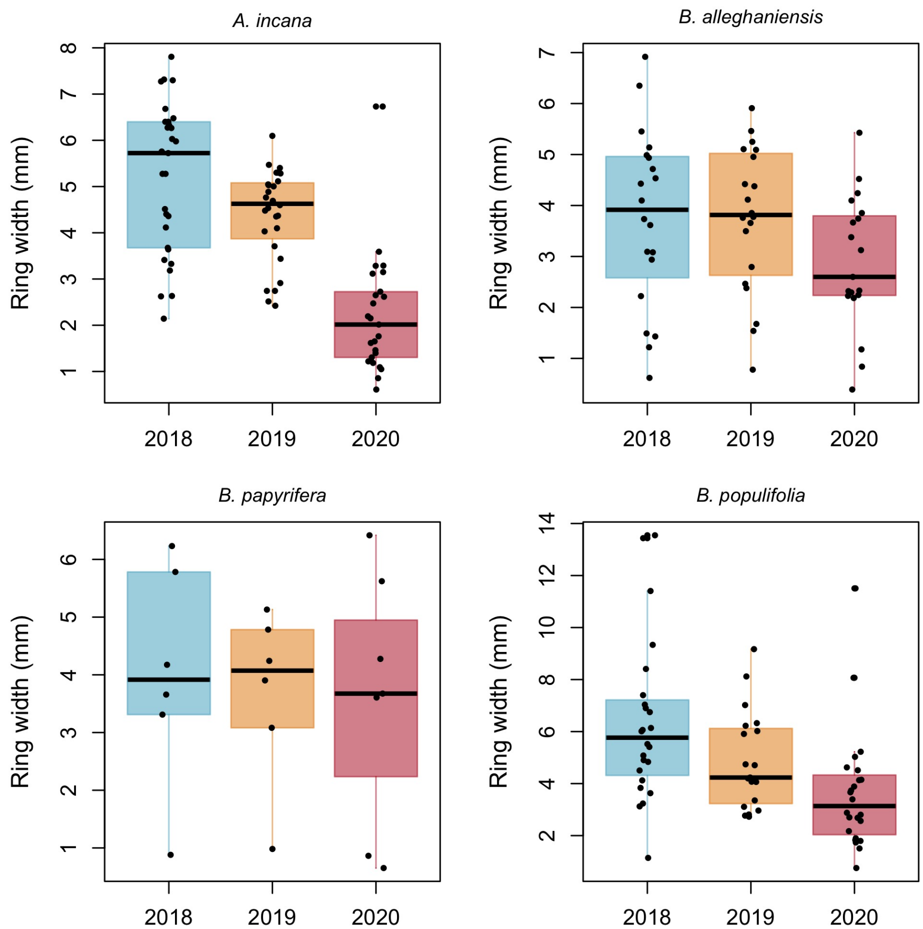

Figure S6: Boxplot representing the empirical data of our ring widths (mm) facetted by species. The dots represent the raw data, the line the mean and the box the IQR.

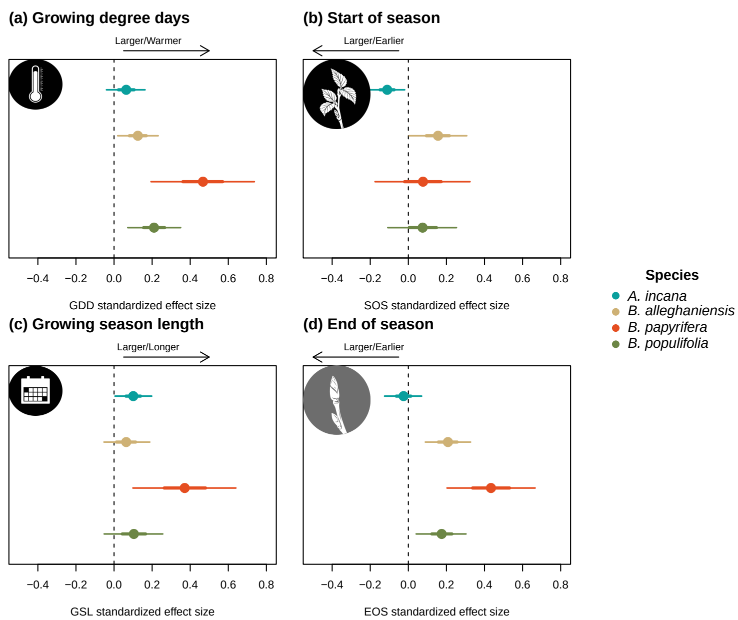

Figure S7: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

Table S3: Posterior means and 90% uncertainty intervals for all model parameters across predictors

Parameter	GDD	GSL	SOS	EOS
$\sigma_{\alpha_{tree}}$	0.17 [0.05, 0.27]	0.18 [0.05, 0.27]	0.21 [0.11, 0.30]	0.18 [0.06, 0.28]
$\sigma_{\alpha_{site}}$	0.18 [0.02, 0.50]	0.18 [0.02, 0.49]	0.16 [0.01, 0.45]	0.18 [0.03, 0.50]
σ_y	0.48 [0.43, 0.52]	0.48 [0.44, 0.53]	0.47 [0.43, 0.52]	0.47 [0.42, 0.52]
$\beta_{spp_{A.incana}}$	0.07 [-0.04, 0.17]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
$\beta_{spp_{B.alleghaniensis}}$	0.13 [0.01, 0.24]	0.03 [-0.02, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
$\beta_{spp_{B.papyrifera}}$	0.44 [0.18, 0.70]	0.18 [0.05, 0.32]	0.05 [-0.10, 0.21]	0.21 [0.09, 0.33]
$\beta_{spp_{B.populifolia}}$	0.21 [0.07, 0.36]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.17]	0.10 [0.02, 0.17]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.03 [-0.22, 0.15]	-0.05 [-0.26, 0.13]	-0.07 [-0.27, 0.09]	-0.04 [-0.24, 0.15]
$\alpha_{site_{HarvardForest(MA)}}$	-0.10 [-0.33, 0.08]	-0.10 [-0.34, 0.08]	-0.05 [-0.26, 0.11]	-0.09 [-0.33, 0.08]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.04 [-0.14, 0.25]	0.03 [-0.16, 0.24]	0.02 [-0.15, 0.21]	0.05 [-0.14, 0.27]
$\alpha_{site_{WhiteMountains(NH)}}$	0.08 [-0.09, 0.31]	0.07 [-0.10, 0.29]	0.08 [-0.07, 0.29]	0.08 [-0.10, 0.30]
$\alpha_{year_{2018}}$	0.19 [-0.76, 1.10]	0.20 [-0.77, 1.17]	0.20 [-0.72, 1.14]	0.23 [-0.71, 1.19]
$\alpha_{year_{2019}}$	0.17 [-0.76, 1.09]	0.07 [-0.89, 1.05]	0.10 [-0.83, 1.05]	0.12 [-0.83, 1.09]
$\alpha_{year_{2020}}$	-0.43 [-1.38, 0.48]	-0.38 [-1.35, 0.60]	-0.39 [-1.31, 0.56]	-0.47 [-1.42, 0.50]
$\alpha_{spp_{A.incana}}$	1.01 [-3.14, 5.18]	0.12 [-4.04, 4.32]	1.58 [-1.98, 5.48]	3.08 [-1.29, 7.36]
$\alpha_{spp_{B.alleghaniensis}}$	0.15 [-3.95, 4.36]	0.34 [-3.82, 4.49]	-1.82 [-5.51, 2.13]	-1.82 [-6.29, 2.64]
$\alpha_{spp_{B.papyrifera}}$	-3.82 [-8.73, 1.09]	-2.67 [-7.34, 1.89]	-0.76 [-4.84, 3.59]	-5.58 [-11.03, -0.04]
$\alpha_{spp_{B.populifolia}}$	-0.69 [-4.89, 3.51]	0.22 [-4.00, 4.45]	-0.42 [-4.24, 3.59]	-0.85 [-5.47, 3.82]

Table S4: Posterior means and 90% uncertainty intervals for all model parameters, when using basal area increment (BAI) as the response variable

Parameter	Estimate
$\sigma_{\alpha_{tree}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{site}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{spp_{A.incana}}$	0.11 [-0.03, 0.25]
$\beta_{spp_{B.alleghaniensis}}$	0.20 [0.02, 0.36]
$\beta_{spp_{B.papyrifera}}$	0.92 [0.49, 1.34]
$\beta_{spp_{B.populifolia}}$	0.41 [0.20, 0.61]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.11 [-0.47, 0.17]
$\alpha_{site_{HarvardForest(MA)}}$	-0.09 [-0.45, 0.19]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.06 [-0.23, 0.42]
$\alpha_{site_{WhiteMountains(NH)}}$	0.15 [-0.11, 0.52]
$\alpha_{year_{2018}}$	-0.36 [-1.32, 0.61]
$\alpha_{year_{2019}}$	0.33 [-0.65, 1.32]
$\alpha_{year_{2020}}$	-0.04 [-1.01, 0.94]
$\alpha_{spp_{A.incana}}$	3.23 [-2.25, 8.55]
$\alpha_{spp_{B.alleghaniensis}}$	2.09 [-3.40, 7.54]
$\alpha_{spp_{B.papyrifera}}$	-7.59 [-14.84, -0.29]
$\alpha_{spp_{B.populifolia}}$	-0.07 [-5.98, 5.57]

Chapter 2

CoringTreespotters

Introduction

1. Climate change shifts phenology

Results

1. We found this and that

Discussion

1. This and that are explained by this

Supplemental results

Bibliography

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[5] Stan Development Team. RStan: the R interface to Stan, 2025. URL https://mc-stan.org/.

Appendix A

Supporting Materials

This would be any supporting material not central to the dissertation. For example:

- additional details of methodology and/or data;
- diagrams of specialized equipment developed.;
- copies of questionnaires and survey instruments.